

EFFECT OF HIGH DILUTIONS OF ACETYLCHOLINE AND ATROPINE ON HEART RATE: A STUDY ON ISOLATED RABBIT'S PERFUSED HEART

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ABSTRACT

Effects of Acetylcholine (ACh) on heart are same as those of vagal stimulation thus; it slows the rate and diminishes the force of beat, which is abolished by Atropine (Atr). It is not known that high dilution (beyond Avagadro's number) of ACh and Atr can produce same pharmacological effect or not, that is reported for other substances (Jerman, *et al.*, 1999) with their simple and succussed dilutions. In the present study, such dilutions (ranging from 10^{-3} to 10^{-36}) of both the drugs have been tested on the isolated Rabbit's heart perfused by using Langendorff assembly. Statistically significant difference has been observed between both series at 10^{-4} (ACh) and 10^{-5} , 10^{-24} and 10^{-36} (Atr+ACh). It is concluded that, high dilution of ACh and Atr, prepared beyond Avagadro's number also possesses imprints of original molecules as reported for other substances. Further, succussion can enhance this process. A chemical identification of such imprints is suggested for pharmacological agents.

INTRODUCTION

Peculiar physiological and pharmacological effects of various agents have already been established in concentrated as well as diluted forms. However, effects of these agents at high dilutions beyond Avagadro's number are not yet investigated. While, various physiological activities of different substances have been observed at different dilutions beyond 10^{-24} , when the original molecule of the substances does not exist in the solution (Boyd 1942 & 1954, Saify *et al.*, 1976, Benveniste 1988, Azeem, 1999, Arifa, *et al.* (2000), Arifa, 2001 and Erum, 2003). Further, succussion (violent shaking) of diluted substances has been reported to differ in their action and structure with the diluted substances that are not succussed, i.e., simple dilutions (Kolisko, 1975 as quoted by Hughes, 2003).

In the light of the above mentioned facts, the present study has been carried out to observe the effects of both series (simple, SD and succussed, SUD) from low to high

dilutions (10^{-3} to 10^{-36}) beyond Avagadro's number for Acetylcholine (ACh) and Atropine (Atr) on isolated Rabbit's Heart.

It is hypothesized that negative chronotropic effect of ACh and its antagonization by Atr may differ on using their SUD instead of SD. Further, high dilutions of both of these kinds, having none of the original drug molecule may show physiological as well as pharmacological effects.

MATERIALS AND METHODS

i) *Experimental Protocol:*

Rabbits (*Oryctolagus cuniculus*) of either sex ranging from 1.0-1.5 Kg were used for experiments. These experimental animals were killed by decapitation for isolation of heart, immediately. The isolated heart was mounted and perfused through Langendorff assembly according to Azeem *et al.* (1999). The perfusate was oxygenated (95% O₂ and 5% CO₂) Krebs's Henseleit Buffer maintained at pH 7.4 and regulated at 37°C. The perfusion

was carried out through the aorta at a constant pressure monitored through mercury manometer. The composition (mM) of oxygenated Kreb's Henseleit Buffer was 118; NaCl, 4.70; KCl, 1.25; CaCl₂, 1.20; MgSO₄, 25; NaHCO₃ and 11; Glucose.

ii) Preparation of Dilution:

In all the experiments, Simple dilutions (SD) and Succussed dilutions (SUD) of ACh and Atr were prepared by dissolving 0.01gm of drug in 10ml distilled/de-ionized water. This dilution (10^{-3}) contains 5.5mM of original drug, from which further dilutions were prepared upto 10^{-36} dilution; this last dilution may contains 5.5×10^{-33} mM of ACh

and Atr i.e., above Avagadro's number. For the preparation of SUD the diluted drugs from 10^{-3} to 10^{-36} were shaken for two hours in a mechanical shaker.

iii) Recording of Cardiac Activity of Isolated Perfused Heart:

The isolated heart was perfused through aorta for coronary perfusion and its ventricular activity was recorded through isotonic transducer on Oscillograph (Harvard apparatus Ltd. UK, Cat no. 50-8622). The mechanical records obtained before and after the infusion of 1 ml of the dilutions (simple and succussed) were later used for the measurement of Heart rate (HR).

Table-1

Statistical comparison between the effect of various simple and succussed dilutions of Acetylcholine on the average values of Heart rate (Calculated as percent of control)

ACh Dilutions	Simple Dilutions		Succussed Dilutions		Significance Level
10^{-3}	68.34 ± 6.45	(17)	73.88 ± 6.01	(17)	P>0.05
10^{-4}	94.48 ± 5.24	(14)	75.81 ± 4.65	(14)	P<0.01
10^{-5}	98.40 ± 3.62	(14)	97.44 ± 0.78	(27)	P>0.05
10^{-6}	*101.25 ± 2.95	(13)	*100.55 ± 1.34	(17)	P>0.05
10^{-7}	94.67 ± 2.23	(15)	96.16 ± 1.30	(14)	P>0.05
10^{-8}	95.78 ± 1.16	(14)	97.55 ± 1.48	(13)	P>0.05
10^{-9}	93.23 ± 2.42	(15)	95.80 ± 2.04	(14)	P>0.05
10^{-10}	98.02 ± 2.26	(16)	98.74 ± 2.85	(17)	P>0.05
10^{-11}	94.22 ± 3.32	(13)	96.37 ± 2.66	(15)	P>0.05
10^{-12}	95.06 ± 1.28	(20)	97.88 ± 2.92	(16)	P>0.05
10^{-15}	96.30 ± 2.08	(16)	97.53 ± 1.33	(11)	P>0.05
10^{-18}	89.12 ± 4.04	(19)	91.11 ± 2.20	(13)	P>0.05
10^{-21}	93.60 ± 2.76	(12)	97.90 ± 1.14	(13)	P>0.05
10^{-24}	92.93 ± 1.65	(22)	96.75 ± 2.34	(23)	P>0.05
10^{-27}	96.83 ± 1.58	(13)	*100.50 ± 4.30	(14)	P>0.05
10^{-30}	98.75 ± 1.75	(13)	93.70 ± 3.34	(13)	P>0.05
10^{-33}	96.86 ± 1.88	(12)	95.21 ± 1.20	(13)	P>0.05
10^{-36}	89.12 ± 3.26	(14)	92.73 ± 2.52	(16)	P>0.05

Reverse effect of ACh

Significant difference

All values are presented as mean ± standard error. While figure in parenthesis represents the number of observations.

Table-2

Statistical comparison between the effect of various simple and succussed dilutions of Acetylcholine in the presence of Atropine on the average values of Heart rate (Calculated as percent of control)

ACh Dilutions	Simple Dilutions		Succussed Dilutions		Significance Level
10^{-3}	**91.02 $\pm$ 3.63	(9)	**91.71 $\pm$ 1.80	(14)	P>0.05
10^{-4}	**96.27 $\pm$ 2.63	(12)	**95.73 $\pm$ 1.38	(21)	P>0.05
10^{-5}	*110.10 $\pm$ 2.53	(8)	**96.22 $\pm$ 0.95	(18)	P<0.0005
10^{-6}	*100.32 $\pm$ 0.44	(7)	**99.77 $\pm$ 1.86	(22)	P>0.05
10^{-7}	**96.23 $\pm$ 2.20	(10)	**94.54 $\pm$ 1.49	(18)	P>0.05
10^{-8}	**96.65 $\pm$ 2.89	(9)	**92.06 $\pm$ 3.47	(18)	P>0.05
10^{-9}	**94.54 $\pm$ 1.43	(15)	**96.06 $\pm$ 1.83	(18)	P>0.05
10^{-10}	**94.71 $\pm$ 1.85	(17)	**95.78 $\pm$ 1.75	(11)	P>0.05
10^{-11}	**95.34 $\pm$ 1.69	(15)	**94.73 $\pm$ 1.81	(11)	P>0.05
10^{-12}	**93.14 $\pm$ 2.11	(14)	**96.92 $\pm$ 3.63	(10)	P>0.05
10^{-15}	**98.97 $\pm$ 1.44	(11)	**93.45 $\pm$ 4.94	(8)	P>0.05
10^{-18}	**90.49 $\pm$ 1.75	(22)	**91.29 $\pm$ 1.40	(18)	P>0.05
10^{-21}	**93.59 $\pm$ 3.22	(14)	**91.29 $\pm$ 3.62	(9)	P>0.05
10^{-24}	**87.54 $\pm$ 5.02	(10)	**68.23 $\pm$ 4.51	(10)	P<0.01
10^{-27}	**92.85 $\pm$ 2.72	(7)	**98.79 $\pm$ 2.23	(7)	P>0.05
10^{-30}	**89.39 $\pm$ 3.69	(11)	**92.23 $\pm$ 2.26	(9)	P>0.05
10^{-33}	**90.43 $\pm$ 3.51	(10)	**82.43 $\pm$ 4.64	(9)	P>0.05
10^{-36}	**91.65 $\pm$ 4.16	(11)	**76.88 $\pm$ 6.86	(9)	P<0.05

* Reverse effect of ACh

** Reverse effect of Atr

Significant difference

All values are presented as mean $\pm$ standard error. While figure in parenthesis represents the number of observations.

iv) Data Analysis:

All measured values of systolic force were used for the calculation of their average and standard error. The statistical comparison was however made between the effect of simple and succussed dilutions by using students t-test at a significance level of 0.05.

RESULTS

Effect of ACh on heart rate

i) Effects of Simple Dilutions:

The SD of Acetylcholine (10^{-3} to 10^{-36}) has been found to decrease the HR at all of its dilutions except 10^{-6} only. This decrease was found maximum i.e., 32% less than control at 10^{-3} . However, at 10^{-6} a negligible increase

was found in the average value of this parameter representing the reverse effect of ACh (Table 1).

ii) Effects of Succussed Dilutions:

The SUD of ACh (10^{-3} to 10^{-36}) have also been found to decrease the HR at all dilutions except 10^{-6} and 10^{-27} (Table-1). This decrease was found maximum, i.e., 27% less than control at 10^{-3} . However, ACh dilutions at 10^{-6} and 10^{-27} have demonstrated a negligible increase in the HR, having its reverse effect.

iii) Comparison between the effects of Simple and Succussed Dilutions:

Statistical comparison between the effects of SD and SUD of ACh on HR demonstrates

significant difference at 10^{-4} ($P < 0.01$) only (Table 1).

Effect of ACh in the Presence of Atr on Heart Rate:

i) Effects of Simple Dilutions:

SD of ACh (10^{-3} to 10^{-36}) used after the administration of respective dilutions of their antagonist Atropine (Atr) have been found to decrease the HR at all of its dilutions except 10^{-5} and 10^{-6} (Table 2). This decrease was found maximum at 10^{-24} being 13% of control and it was the reverse (opposite to the normal) effect of Atr. While, at 10^{-5} and 10^{-6} dilution of ACh the HR was slightly higher than control respectively being the reverse effect of ACh. This reverse effect represents that Atr did not antagonize the effect of ACh at this dilution.

ii) Effects of Succussed Dilutions:

The SUD of ACh after the administration of respective antagonist Atr have been found to decrease the HR at all dilutions ranging from 10^{-3} to 10^{-36} . This decrease was found

maximum at 10^{-24} having 32% less than control, thus representing the reverse effect of Atr (Table-2).

iii) Comparison between the effects of Simple and Succussed Dilutions:

Statistical comparison between the effects of SD and SUD of ACh after the administration of its antagonist Atr demonstrates the significant difference in the HR at 10^{-5} ($P < 0.0005$), 10^{-24} ($P < 0.01$) and 10^{-36} ($P < 0.05$) only as shown in Table-2.

DISCUSSION

Effect of SD and SUD of ACh:

The negative chronotropic effect was found highly pronounced at lower dilutions, but it become negligible, at 10^{-5} SD and SUD (Fig.1). It is suggested that both the SD and SUD have produced specific but unstable molecular changes in the parent drug when interacted with water during the preparation of dilutions. However, it is suggested that the

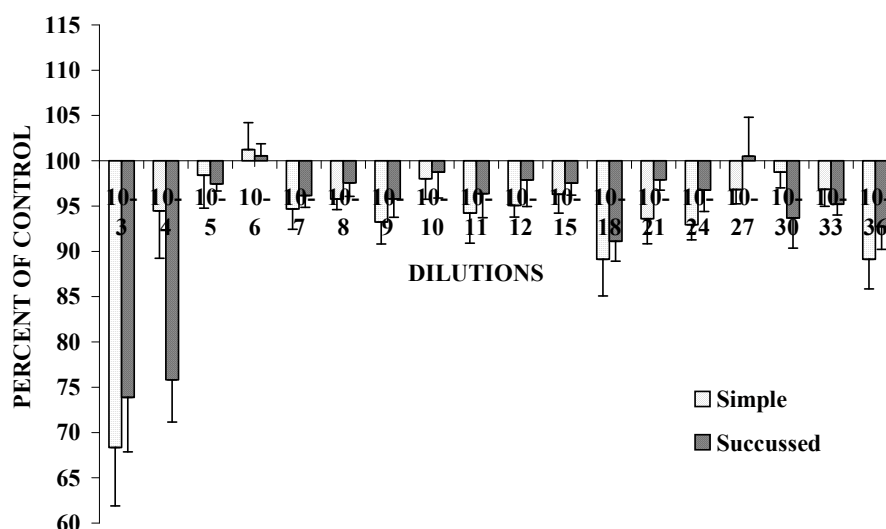


Fig. 1: Comparative Effect of Simple and Succussed Dilutions of Acetylcholine on Heart Rate

fluctuations in the negative chronotropic effect observed for different dilutions of ACh, might be linked with fluctuated increase in ACh-dependent K^+ conductance as already reported in atria and ventricular cells by Sakmann *et al.*, (1983), Wang & Lipisius, (1995) and Demirl *et al.* (1999).

However, cholinergic action of ACh, at 10^{-6} (SD & SUD) as well as at 10^{-27} (SUD) indicate an expected observation that lead the thought to consider possible changes in drug molecule (Hughes, 2003) showing reverse physiological effect of ACh, i.e., positive chronotropic effect. This effect is however, negligible but cannot be ignored.

Later, the ACh SD and SUD dilutions (10^{-18} to 10^{-36}) had produced potentized negative chronotropic effect than the earlier dilutions (10^{-7} to 10^{-15}) used. Such changes in the action of ACh clearly indicate some kind of drug water interaction.

Effect of SD and SUD OF ACh in the Presence of Atr:

Lower than control values of HR (negative chronotropic effect) were also observed at all dilutions of ACh in the presence of Atr, except, 10^{-5} and 10^{-6} (SD). This effect was specifically pronounced at higher dilutions, especially 10^{-24} , 10^{-33} and 10^{-36} (Fig. 2). It is therefore clear that earlier report for some kind of physico-chemical change at higher dilutions is true. Such change has potentized the parent drug molecules (Boyd, 1954, Saify *et al.*, 1976 and Benveniste, 1988) that have resulted in a prominent cholinomimetic effect in the presence of Atr. That was not observed at low dilutions (Fig. 2). It is therefore clear that some kind of molecular change produced on dilutions has prevented the Atr from its action and the ACh effect become prominent which is also evident in Fig.1 at 10^{-24} , but of lesser intensity. However, the inability of Atr to antagonize ACh in the present study indicates that I_k ACh channels (Carmeliet and Mubagwa,

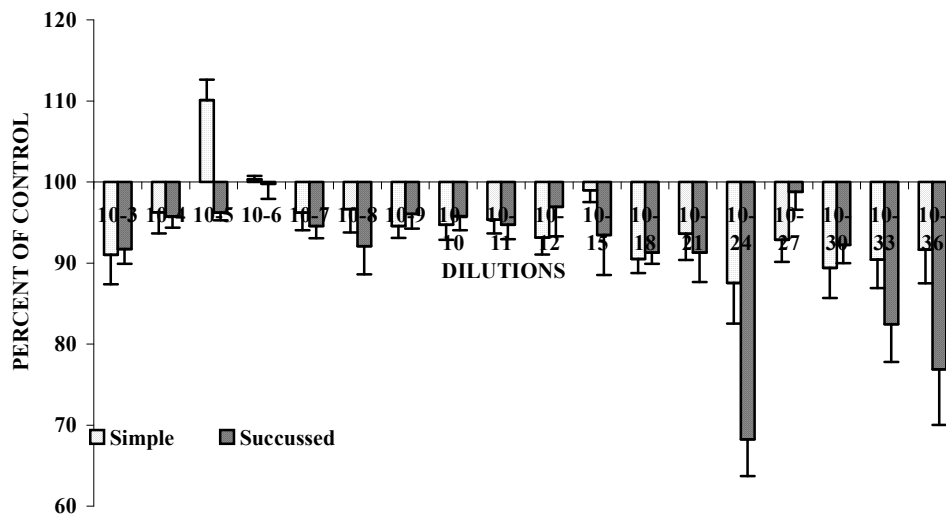


Fig. 2: Comparative Effect of Simple and Succussed Dilutions of Acetylcholine in the Presence of Atropine on Heart Rate.

1986) are probably not blocked by Atr (Brown, 1991) in the present study at higher dilutions.

On the other hand negligible positive chronotropic effect observed at 10^{-5} and 10^{-6} (SD) also suggest mysterious effect of dilutions that cannot be explained on the basis of this study, unless physico-chemical studies are carried out.

CONCLUSION

Three kinds of changes are evident:

- i) Reduction in negative chronotropic effect upto 10^{-6} SD and SUD.
- ii) Revival of negative chronotropic effect beyond 10^{-7} that continued till 10-36.
- iii) Negligible positive chronotropic effect observed at 10^{-5} , 10^{-6} and 10^{-27} .

It is concluded that drug water interaction do offer changes (though not stable) in the parent drug molecules and their imprints in water exists beyond 10^{-24} dilutions. These imprints are probably responsible for physiological and pharmacological effects at high dilutions.

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